

Part - A : Basic Pattern Printing

1. Print Square Star Pattern

```
#include <stdio.h>
int main()
{
    int n = 4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
* * * *
* * * *
* * * *
* * * *
```

2. Print Left Triangle Pattern

```
#include <stdio.h>

int main()
{
    int n=5;
    for(int i=0 ; i<n; i++){
        for(int j=0 ; j<=i; j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
*
* *
* * *
* * * *
* * * * *
```

3. Print Inverted Left Triangle

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=n; i>=1; i--){
        for(int j=1; j<=i; j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
* * * * *
* * * *
* * *
* *
*
```

4. Print Number Triangle

```
#include <stdio.h>
int main ()
{
    int n=5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf ("%d", j);
        }
        printf ("\n");
    }
    return 0;
}
```

OUTPUT:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```


5. Print Same Number Triangle

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n=5;
```

```
    for (int i=1; i<=n; i++) {
```

```
        for (int j=1; j<=i; j++) {
```

```
            printf("%d", i);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

OUTPUT:

1

2 2

3 3 3

4 4 4 4

5 5 5 5 5

Part-B : Intermediate Pattern Printing

6. Print Right Aligned Triangle

```
#include <stdio.h>
int main()
{
    int n=5;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i; j++) {
            printf(" ");
        }
        for (int k=1; k<=i; k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
      *
     **
    ***
   ****
  *****
```

7. Print Inverted Right Triangle

```
#include <stdio.h>
int main()
{
    int n = 5;
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (int k = 1; k <= i; k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
* * * * *
 * * * *
  * * *
   * *
    *
```

8. Print Pyramid Pattern

```
#include <stdio.h>
int main ()
{
    int n=5;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i; j++) {
            printf (" ");
        }
        for (int k=1; k<=2*i-1; k++) {
            printf ("*");
        }
        printf ("\n");
    }
    return 0;
}
```

OUTPUT

```
      *
     **
    ***
   ****
  *****
 *****
*****
```


9. Print Inverted Pyramid

```
#include <stdio.h>
int main ()
{
    int n = 5;
    for (int i = n; i >= 1; i--) {
for (int j = 1; j <= n-i; j++) {
        for (int j = 1; j <= n-i; j++) {
            printf(" ");
        }
        for (int k = 1; k <= 2*i - 1; k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
* * * * *
 * * * *
  * * *
   * *
    *
```


10. Print Hollow Square

```
#include <stdio.h>
int main()
{
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (i == 1 || j == 1 || i == n || j == n) {
                printf ("*");
            } else {
                printf (" ");
            }
        }
        printf ("\n");
    }
    return 0;
}
```

OUTPUT:

```
* * * * *
*           *
*           *
*           *
* * * * *
```

Part-C: Mixed Patterns (if-else + loop)

11. Hollow Left Triangle

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=1; i<=n; i++){
        for(int j=1; j<=i; j++){
            if (j==1 || j==i || i==n) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
*
* *
*  *
*   *
*    *
*   * * *
```

12. Number Pattern (Even / Odd)

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=1; i<=n; i++){
        for(int j=1; j<=i; j++){
            if(j%2 == 0){
                printf("E");
            } else {
                printf("O");
            }
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
O
O E
O E O
O E O E
O E O E O
```

13. Floyd's Triangle

```
#include <stdio.h>
int main()
{
    int n=5, num=1;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=i; j++) {
            printf("%d", num);
            num++;
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```


14. Reverse Number Triangle

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=1; i<=n; i++){
        for(int j=1; j<=i; j++){
            printf("%d", n-j+1);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
5
5 4
5 4 3
5 4 3 2
5 4 3 2 1
```

15. Binary Pattern

```
#include <stdio.h>
```

```
int main()
```

```
{ int n=5, num = 1;
```

```
  for(int i = 1; i <= n; i++) {
```

```
    for(int j = 1; j <= i; j++) {
```

```
      printf("%d", num);
```

```
      if(num == 1) {
```

```
        num = 0;
```

```
      } else {
```

```
        num = 1;
```

```
      }
```

```
    }
```

```
    printf("\n");
```

```
  }
```

```
  return 0;
```

```
}
```

OUTPUT

```
1
0 1
0 1 0
1 0 1 0
1 0 1 0 1
```

16. Pattern with Condition (Divisible by 3)

```
#include <stdio.h>
int main()
{
    int n=5;
    for (int i=1; i<=n; i++){
        for (int j=1; j<=i; j++){
            if (j%3==0){
                printf (" *");
            } else {
                printf ("%d", j);
            }
        }
        printf ("\n");
    }
    return 0;
}
```

OUTPUT:

```
1
1 2
1 2 *
1 2 * 4
1 2 * 4 5
```

17. Character Pattern

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=1; i<=n; i++){
        for(int j=0; j<i; j++){
            printf("%c", 'A'+j);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT

```
A
A B
A B C
A B C D
A B C D E
```


18. Reverse Character Pattern

```
#include <stdio.h>
int main()
{
    int n=5;
    for (int i=1; i<=n; i++) {
        for (int j=0; j<n; j++) {
            printf("%c", 'A'+n-j-1);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

E

E D

E D C

E D C B

E D C B A

19. Alternating Star Pattern

```
#include <stdio.h>
int main()
{
    int n=5;
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=n; j++) {
            printf(" * ");
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

```
*
* *
* * *
* * * *
* * * * *
```

20. Multiplication Pattern

```
#include <stdio.h>

int main ()
{
    int n=5;
    for (int i=1; i<=n; i++){
        for (int j=1; j<=i; j++){
            printf ("%d", i*j);
        }
        printf ("\n");
    }
    return 0;
}
```

OUTPUT

```
1
2 4
3 6 9
4 8 12 16
5 10 15 20 25
```

Part-D: Output Tracing with Simulation

21. Output Tracing - 1

Simulation

$i = 1 \mid 1 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 1 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 2 \mid 2 \leq 1 \text{ (f)} \rightarrow \text{newline}$

$i = 2 \mid 2 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 2 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 2 \mid 2 \leq 2 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 3 \mid 3 \leq 2 \text{ (f)} \rightarrow \text{newline}$

$i = 3 \mid 3 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 3 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 2 \mid 2 \leq 3 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 3 \mid 3 \leq 3 \text{ (t)} \rightarrow \text{print} \rightarrow *$

$j = 4 \mid 4 \leq 3 \text{ (f)} \rightarrow \text{newline}$

$i = 4 \mid 4 \leq 3 \text{ (f)} \text{ [Loop End]}$

OUTPUT:

*

* *

* * *

22. Output Tracing - 2

Simulation

$i = 1 \mid 1 \leq 4 \text{ (t)}$

$j = 4 \mid 4 \geq 1 \text{ (t)} \rightarrow \text{print} \rightarrow 4$

$j = 3 \mid 3 \geq 1 \text{ (t)} \rightarrow \text{print} \rightarrow 3$

$j = 2 \mid 2 \geq 1 \text{ (t)} \rightarrow \text{print} \rightarrow 2$

$j = 1 \mid 1 \geq 1 \text{ (t)} \rightarrow \text{print} \rightarrow 1$

$j = 0 \mid 0 \geq 1 \text{ (f)} \rightarrow \text{newline}$

$i = 2 \mid 2 \leq 4 \text{ (t)}$

$j = 4 \mid 4 \geq 2 \text{ (t)} \rightarrow \text{print} \rightarrow 4$

$j = 3 \mid 3 \geq 2 \text{ (t)} \rightarrow \text{print} \rightarrow 3$

$j = 2 \mid 2 \geq 2 \text{ (t)} \rightarrow \text{print} \rightarrow 2$

$j = 1 \mid 1 \geq 2 \text{ (f)} \rightarrow \text{newline}$

$i = 3 \mid 3 \leq 4 \text{ (t)}$

$j = 4 \mid 4 \geq 3 \text{ (t)} \rightarrow \text{print} \rightarrow 4$

$j = 3 \mid 3 \geq 3 \text{ (t)} \rightarrow \text{print} \rightarrow 3$

$j = 2 \mid 2 \geq 2 \text{ (t)} \rightarrow \text{newline}$

$i = 4 \mid 4 \leq 4 (t)$

$j = 4 \mid 4 > 4 (t) \rightarrow \text{print} \rightarrow 4$

$j = 3 \mid 3 > 4 (f) \rightarrow \text{print} \rightarrow \text{newline}$

$i = 5 \mid 5 \leq 4 (f) \quad [\text{Loop End}]$

OUTPUT:

4 3 2 1

4 3 2

4 3

4

23. Output Tracing - 3

Simulation

$i = 1 \mid 1 \leq 3 (t)$

$j = 1 \mid 1 \leq 3 (t) \mid 1 == 1 (t) \rightarrow \text{print} \rightarrow 1$

$j = 2 \mid 2 \leq 3 (t) \mid 1 == 2 (f) \rightarrow \text{print} \rightarrow 0$

$j = 3 \mid 3 \leq 3 (t) \mid 1 == 3 (f) \rightarrow \text{print} \rightarrow 0$

$j = 4 \mid 4 \leq 3 (f) \rightarrow \text{print} \rightarrow \text{newline}$

$i = 2 \mid 2 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 3 \text{ (t)} \mid 2 == 1 \text{ (f)} \rightarrow \text{print} \rightarrow 0$

$j = 2 \mid 2 \leq 3 \text{ (t)} \mid 2 == 2 \text{ (t)} \rightarrow \text{print} \rightarrow 1$

$j = 3 \mid 3 \leq 3 \text{ (t)} \mid 2 == 3 \text{ (f)} \rightarrow \text{print} \rightarrow 0$

$j = 4 \mid 4 \leq 3 \text{ (f)} \rightarrow \text{print} \rightarrow \text{newline}$

$i = 3 \mid 3 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 3 \text{ (t)} \mid 3 == 1 \text{ (f)} \rightarrow \text{print} \rightarrow 0$

$j = 2 \mid 2 \leq 3 \text{ (t)} \mid 3 == 2 \text{ (f)} \rightarrow \text{print} \rightarrow 0$

$j = 3 \mid 3 \leq 3 \text{ (t)} \mid 3 == 3 \text{ (t)} \rightarrow \text{print} \rightarrow 1$

$j = 4 \mid 4 \leq 3 \text{ (f)} \rightarrow \text{print} \rightarrow \text{newline}$

$i = 4 \mid 4 \leq 3 \text{ (f)} \quad [\text{Loop End}]$

OUTPUT

1 0 0

0 1 0

0 0 1

24. Output Tracing - 4

Simulation

$$i = 3 \mid 3 \geq 1 (t)$$

$$j = 1 \mid 1 \leq 3 (t) \rightarrow \text{print} \rightarrow 3$$

$$j = 2 \mid 2 \leq 3 (t) \rightarrow \text{print} \rightarrow 3$$

$$j = 3 \mid 3 \leq 3 (t) \rightarrow \text{print} \rightarrow 3$$

$$j = 4 \mid 4 \leq 3 (f) \rightarrow \text{print} \rightarrow \text{newline}$$

$$i = 2 \mid 2 \geq 1 (t)$$

$$j = 1 \mid 1 \leq 2 (t) \rightarrow \text{print} \rightarrow 2$$

$$j = 2 \mid 2 \leq 2 (t) \rightarrow \text{print} \rightarrow 2$$

$$j = 3 \mid 3 \leq 2 (f) \rightarrow \text{print} \rightarrow \text{newline}$$

$$i = 1 \mid 1 \geq 1 (t)$$

$$j = 1 \mid 1 \leq 1 (t) \rightarrow \text{print} \rightarrow 1$$

$$j = 2 \mid 2 \leq 1 (f) \rightarrow \text{print} \rightarrow \text{newline}$$

$$i = 0 \mid 0 \geq 1 (f) \quad [\text{Loop Ends}]$$

Output:

3 3 3

2 2

1

25. Output Tracing - 5

Simulation

$i = 1 \mid 1 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 1 \text{ (t)}$

$x = 2 \mid 2 \cdot 2 \neq 0 \text{ (t)} \rightarrow \text{print} \rightarrow 2$

$j = 2 \mid 2 \leq 2 \text{ (f)} \rightarrow \text{print} \rightarrow \text{newline}$

$i = 2 \mid 2 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 1 \text{ (t)}$

$x = 3 \mid 3 \cdot 2 \neq 0 \text{ (f)} \rightarrow \text{print} \rightarrow *$

$j = 2 \mid 2 \leq 2 \text{ (t)}$

$x = 4 \mid 4 \cdot 2 \neq 0 \text{ (t)} \rightarrow \text{print} \rightarrow 4$

$j = 3 \mid 3 \leq 2 \text{ (f)} \rightarrow \text{print} \rightarrow \text{newline}$

$i = 3 \mid 3 \leq 3 \text{ (t)}$

$j = 1 \mid 1 \leq 3 \text{ (t)}$

$x = 5 \mid 5 \cdot 2 \neq 0 \text{ (f)} \rightarrow \text{print} \rightarrow *$

$j = 2 \mid 2 \leq 3 \text{ (t)}$

$x = 6 \mid 6 \cdot 2 \neq 0 \text{ (t)} \rightarrow \text{print} \rightarrow 6$

$j = 3 \mid 3 \leq 3 \text{ (t)}$

$x = 7 \mid 7 \cdot 2 = 20 \text{ (f)} \rightarrow \text{print} \rightarrow *$

$j = 4 \mid 4 \leq 3 \text{ (f)} \rightarrow \text{print} \rightarrow \text{newline}$

$i = 4 \mid 4 \leq 3 \text{ (f)} \quad [\text{Loop End}]$

OUTPUT

2

* 4

* 6 *